

35. DEVELOPMENT OF THE COLIC FRAME : THE DUODENUM

DURATION : 28:04

STUDY SHEET

COURSE SUMMARY

This video teaches about the development of the duodenum in relation to the liver, pancreas, and sphincters, which are essential for digestion and homeostasis. By exploring the 90-degree rotational dynamics, the interconnection of organs, and the impact of movements on hormonal and neurovascular balance, the speaker highlights how an imbalance in the duodenal framework can influence not only nutrient absorption but also fertility. The importance of sphincters, their role as joints, and their mechanical functionality are analysed, while also emphasising their impact on the immune system and emotional state.

DEVELOPMENT OF THE COLIC FRAME: THE DUODENUM

This phase of **duodenal** development is characterised by a **rotation** and **organisation** of the duodenal frame, small intestine, and colon. We will explore this dynamic in relation to the duodenum.

a) *Situation et fixité* (fig. 68). — Le duodénum est très profond, central, devant le rachis de L1 à L4.

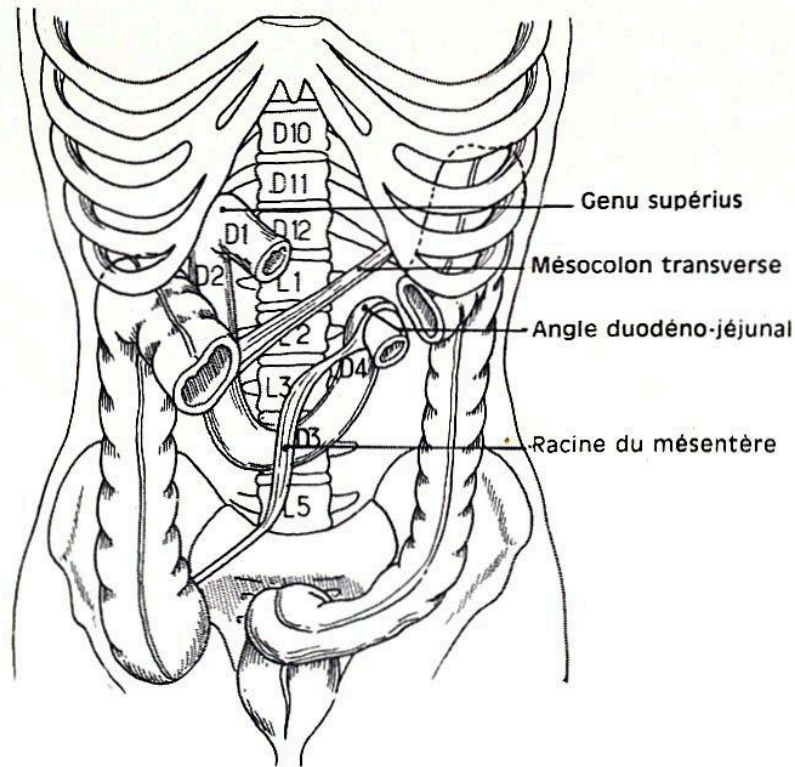


FIG. — Duodenum and mesentery

The **duodenum**, small intestine, and colon develop in a **spiral** movement, where the **liver** plays a driving role. This development is organised around a **facial pentagon** centred on the **pancreas**. The lesser sac, with the right leaflet of the visceral peritoneum at the level of the stomach, creates a fold which, thanks to the development of the liver, leads to a dynamic where the stomach positions itself: the right part becomes posterior and the left part anterior.

During this phase, the duodenum undergoes a **90-degree rotation** from left to right, which constitutes an axis of rotation for **hepatopancreatic**, gastric, and splenic organisation. Initially, the duodenum is a straight tube which, as it develops, creates **spaces for movement** and **torsion**. These rotations generate areas called **sphincters**, which function like joints. For example, the sphincter between the stomach and the duodenum, called the **pyloric sphincter**, is influenced by the movement of the liver and the gastric rotation axis.

The tissue that develops around this dynamic is the **lesser omentum**, which organises and rebalances the pyloric sphincter. The entirety of this movement creates the **sphincteric functions**. In parallel, two pancreatic buds form, one ventral and one dorsal, following the same phase of rotation.

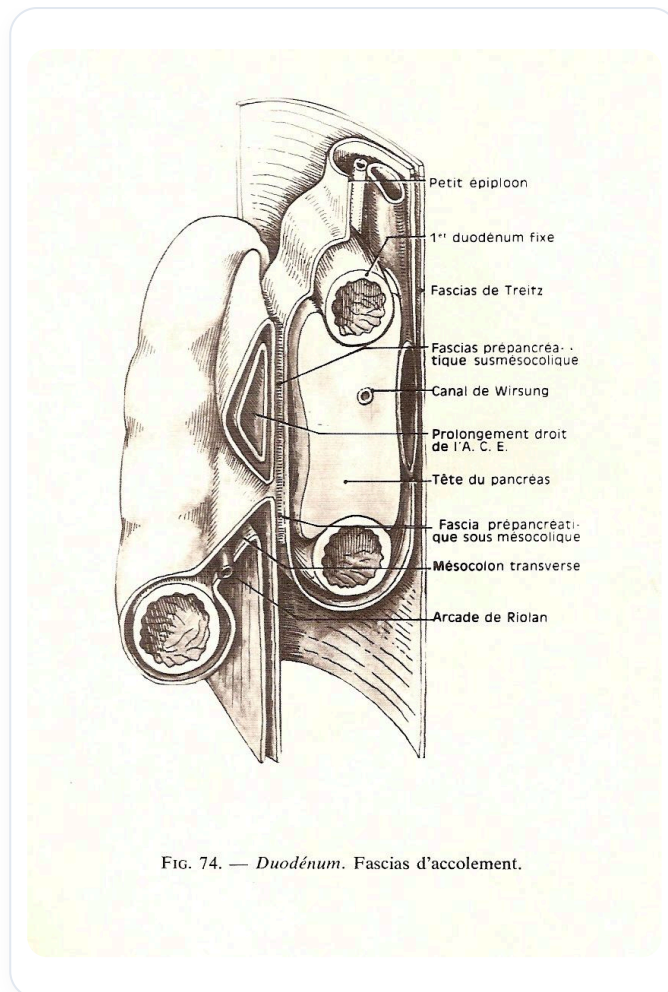


FIG. — Duodenal fascias of fusion

The framework of development at the peritoneal level is a movement between the stomach, the liver, and the colic frame, which organises the sphincteric functions. The pyloric sphincter is created by the movement of the liver and stomach, while the 2D sphincter results from the rotation of the colon and the pancreatic complex. Releasing a 2D sphincter in its embryonic fulcrum allows for the release of the pancreatic facial pentagon and the liver, thereby creating an enzymatic, hormonal, biliary, and neurovegetative reflex zone.

On a hormonal level, there is a parathyroid relationship with calcium receptors, connecting the liver and the stomach. The 2D sphincter balances the development of the pancreas and the liver, while the duodenal sphincter is linked to the small intestine, forming a joint between the duodenocolic system and the mesenteric system.

Information at this level can be modified by gastro-ileal, colo-ileal, or colo-gastric signals. This point becomes crucial for the influx of digestive vascularisation. The **ligament of Treitz** plays a key role, allowing for the increase or decrease of information depending on the caloric state of the food bolus, thus influencing vascular behaviour.

The ileocecal sphincter also develops, linked to the establishment of the frames between the mesentery and the right colic angle. The position of the sphincter can influence the immune system, and a destabilisation of the duodenal frame can affect fertility by disrupting blood flow.

Each sphincter has two main functions: a mechanical function to change the enzymatic, hormonal, and neurovegetative plane, and a neurovascular information function. The muscular plane of the duodenum is a splanchnic pattern, influenced by the diaphragm and the aortic system.

The duodenum, liver, and the entire digestive system are interconnected, forming a complex network of energy and information. Rotational and developmental movements influence the digestive and vegetative state, and a destabilisation of the duodenal frame can lead to disorders such as depression.

The duodenum is a concentrated site of metabolic information, where malabsorption of certain proteins and vitamins can occur. For example, vitamin D, essential for calcium absorption, is linked to the state of the sphincter of D. Osteoporosis can thus have a digestive origin, linked to a hormonal imbalance between the parathyroid and duodenal receptors.

The sphincter of D, in relation to the development of the liver and pancreas, is an essential fixed point. The motility of the liver is crucial for the proper functioning of the digestive system. A destabilisation of the duodenal frame can also disrupt fertility by affecting the ovarian and spermatic arteries.

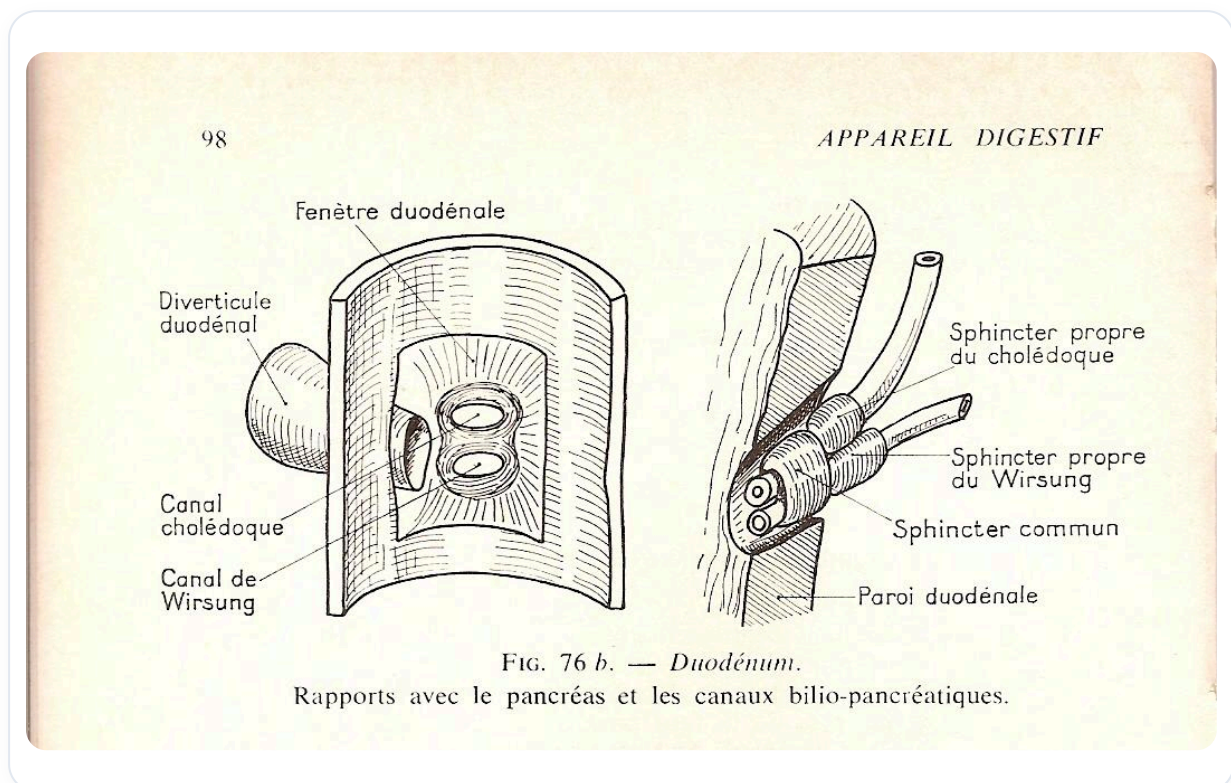


FIG. — Ampulla of Vater

The ligament of Treitz, as a vascular crossroads, influences intra-peritoneal and supramesocolic blood flow. Viscerospasms can have repercussions on blood circulation, thus affecting the metabolic and digestive state.

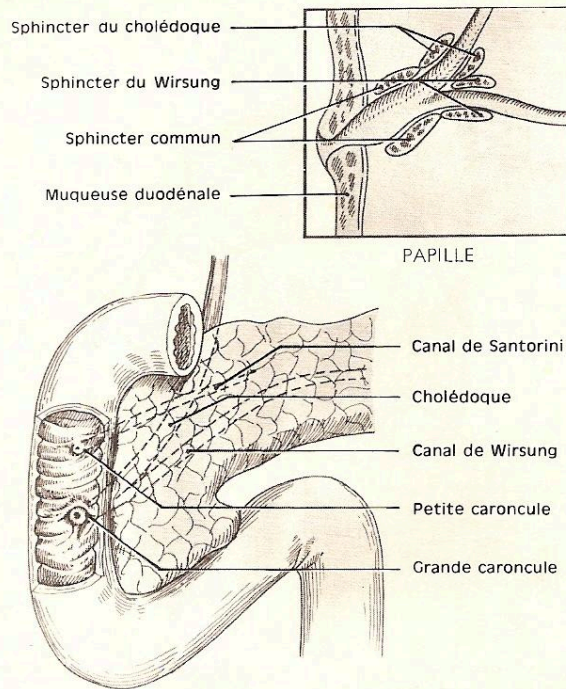


FIG. 77 a. — *Duodénum.*
Terminaison des canaux bilio-pancréatiques

FIG. — Termination of bilio-pancreatic ducts

The treatment of imbalances at the duodenal level requires a holistic approach, taking into account the interactions between different systems. Understanding embryology, anatomy, and physiology is essential to grasp the links between the body, mind, and overall health.

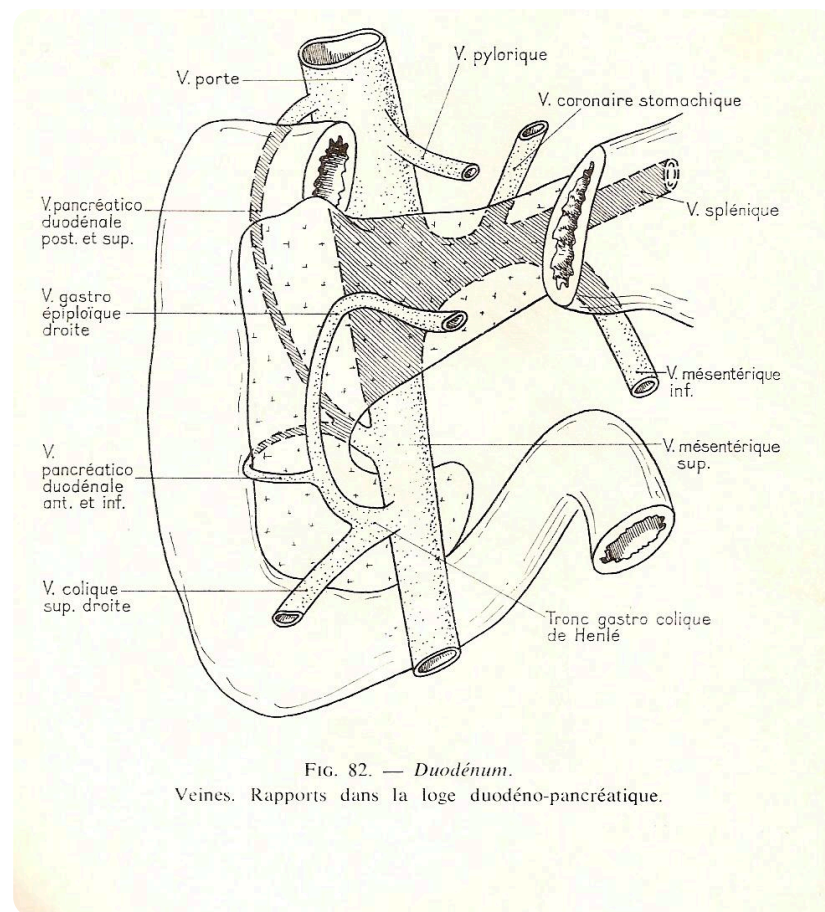


FIG. 82. — *Duodénum*.
Veines. Rapports dans la loge duodéno-pancréatique.

FIG. — Duodenal and pancreatic vascularization

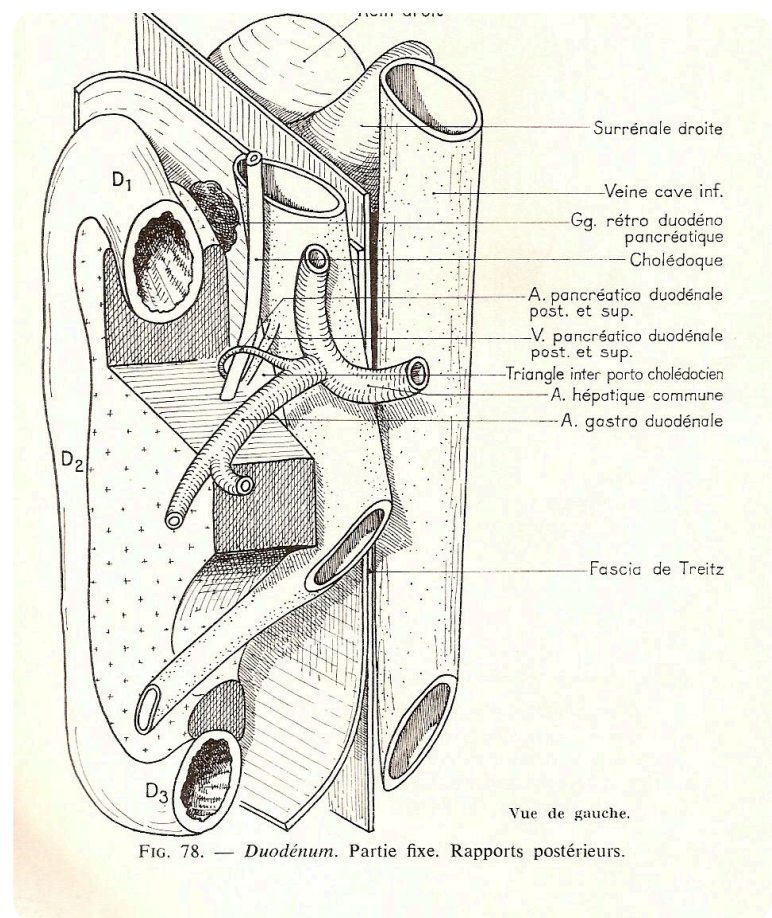


FIG. — *Duodenum posterior relations*